Source folder: /Users/jonat/Documents/GitHub/master_thesis
Files in this part: 17

File: verify_doc.py

```
import docx

def verify(doc_path):
    doc = docx.Document(doc_path)
    print(f"Total paragraphs: {len(doc.paragraphs)}")

    print("\n--- Last 20 Paragraphs ---")
    for i, p in enumerate(doc.paragraphs[-20:]):
        print(f"[{len(doc.paragraphs)-20+i}] {repr(p.text)}")

    print("\n--- Captions Found ---")
    count = 0
    for i, p in enumerate(doc.paragraphs):
        text = p.text.strip()
        if text.lower().startswith("figure"):
             print(f"[{i}] {text[:50]}")
             count += 1

    print(f"Total lines starting with 'Figure': {count}")

if __name__ == "__main__":
    verify("Empirical Analysis of Accuracy.docx")
```

File: verify_results.py

```
"""
Comprehensive Results Verification Report

This script generates a detailed comparison report between
existing results and published results in EXPERIMENT_STATUS.md
"""

import json
import numpy as np
from pathlib import Path
from typing import Dict, List
from datetime import datetime


def load_experiment_results(results_dir: str = "results") -> Dict:
    """Load all experiment results from JSON files."""
    results = {}
    results_path = Path(results_dir)

    for json_file in results_path.glob("*.json"):
        if json_file.name == "attack_evaluation_summary.json":
            continue

        with open(json_file) as f:
            data = json.load(f)
            results[json_file.stem] = data

    return results


def extract_all_metrics(results: Dict) -> Dict:
    """Extract comprehensive metrics from experiment results."""
    metrics = {}

    # Group by configuration
    configurations = {}

    for exp_id, data in results.items():
        if exp_id == "centralized_baseline":
            metrics["centralized"] = {
                "NDCG@10": data.get("final_metrics", {}).get("NDCG@10", 0),
                "Hit@10": data.get("final_metrics", {}).get("Hit@10", 0),
                "MSE": data.get("final_metrics", {}).get("MSE", 0),
```

```
                "MAE": data.get("final_metrics", {}).get("MAE", 0),
            }
            continue

        # Parse experiment ID
        parts = exp_id.split("_seed_")
        if len(parts) == 2:
            config_key = parts[0]
            seed = int(parts[1])
        else:
            continue

        if config_key not in configurations:
            configurations[config_key] = []

        configurations[config_key].append({
            "seed": seed,
            "NDCG@10": data.get("final_metrics", {}).get("NDCG@10", 0),
            "Hit@10": data.get("final_metrics", {}).get("Hit@10", 0),
            "MSE": data.get("final_metrics", {}).get("MSE", 0),
            "MAE": data.get("final_metrics", {}).get("MAE", 0),
            "final_loss": data.get("final_metrics", {}).get("training_loss", 0),
        })

    # Compute statistics across seeds
    for config_key, runs in configurations.items():
        ndcg_values = [r["NDCG@10"] for r in runs]
        hit_values = [r["Hit@10"] for r in runs]
        mse_values = [r["MSE"] for r in runs]
        mae_values = [r["MAE"] for r in runs]

        metrics[config_key] = {
            "NDCG@10": {
                "mean": np.mean(ndcg_values),
                "std": np.std(ndcg_values),
                "min": np.min(ndcg_values),
                "max": np.max(ndcg_values),
                "values": ndcg_values
            },
            "Hit@10": {
                "mean": np.mean(hit_values),
                "std": np.std(hit_values),
                "min": np.min(hit_values),
                "max": np.max(hit_values),
                "values": hit_values
            },
            "MSE": {
                "mean": np.mean(mse_values),
                "std": np.std(mse_values),
            },
            "MAE": {
                "mean": np.mean(mae_values),
                "std": np.std(mae_values),
            },
            "num_runs": len(runs),
            "seeds": [r["seed"] for r in runs]
        }

    return metrics


def generate_report():
    """Generate comprehensive verification report."""
    print("=" * 80)
    print("EXPERIMENT RESULTS VERIFICATION REPORT")
    print(f"Generated: {datetime.now().strftime('%Y-%m-%d %H:%M:%S')}")
    print("=" * 80)

    # Published results from EXPERIMENT_STATUS.md
    published = {
        "centralized": {
            "NDCG@10": 0.2250,
            "Hit@10": 0.3800
        },
```

```
        "dp_inf_alpha_0.5_dim_64_clients_100": {
            "NDCG@10": {"mean": 0.0539, "std": 0.0108},
            "Hit@10": {"mean": 0.0633, "std": 0.0205}
        },
        "dp_8_alpha_0.5_dim_64_clients_100": {
            "NDCG@10": {"mean": 0.0534, "std": 0.0131},
            "Hit@10": {"mean": 0.0600, "std": 0.0082}
        },
        "dp_4_alpha_0.5_dim_64_clients_100": {
            "NDCG@10": {"mean": 0.0479, "std": 0.0091},
            "Hit@10": {"mean": 0.0600, "std": 0.0082}
        },
        "dp_2_alpha_0.5_dim_64_clients_100": {
            "NDCG@10": {"mean": 0.0456, "std": 0.0095},
            "Hit@10": {"mean": 0.0567, "std": 0.0094}
        },
        "dp_1_alpha_0.5_dim_64_clients_100": {
            "NDCG@10": {"mean": 0.0467, "std": 0.0121},
            "Hit@10": {"mean": 0.0533, "std": 0.0047}
        },
        "dp_inf_alpha_0.1_dim_64_clients_100": {
            "NDCG@10": {"mean": 0.0538, "std": 0.0108},
            "Hit@10": {"mean": 0.0633, "std": 0.0205}
        },
        "dp_inf_alpha_1.0_dim_64_clients_100": {
            "NDCG@10": {"mean": 0.0539, "std": 0.0108},
            "Hit@10": {"mean": 0.0633, "std": 0.0205}
        }
    }

    # Load current results
    print("\nLoading experiment results from results/...")
    current_results_raw = load_experiment_results("results")
    current_results = extract_all_metrics(current_results_raw)
    print(f"Loaded {len(current_results)} configurations")

    # Section 1: DP Sweep Results (RQ1)
    print("\n" + "=" * 80)
    print("SECTION 1: DP SWEEP RESULTS (RQ1: Accuracy-Privacy Trade-offs)")
    print("=" * 80)
    print("\nConfiguration: 100 clients, 10 rounds, 3 local epochs, α=0.5")
    print("\n{:<20} {:<25} {:<25} {:<10}".format(
        "DP Budget (ε)", "NDCG@10", "Hit@10", "Status"))
    print("-" * 80)

    dp_configs = [
        ("inf", "dp_inf_alpha_0.5_dim_64_clients_100"),
        ("8", "dp_8_alpha_0.5_dim_64_clients_100"),
        ("4", "dp_4_alpha_0.5_dim_64_clients_100"),
        ("2", "dp_2_alpha_0.5_dim_64_clients_100"),
        ("1", "dp_1_alpha_0.5_dim_64_clients_100"),
    ]

    all_match = True
    for epsilon, config_key in dp_configs:
        if config_key in current_results and config_key in published:
            curr = current_results[config_key]
            pub = published[config_key]

            curr_ndcg = curr["NDCG@10"]["mean"]
            pub_ndcg = pub["NDCG@10"]["mean"]
            curr_hit = curr["Hit@10"]["mean"]
            pub_hit = pub["Hit@10"]["mean"]

            ndcg_match = abs(curr_ndcg - pub_ndcg) < 0.0001
            hit_match = abs(curr_hit - pub_hit) < 0.0001

            status = "■" if (ndcg_match and hit_match) else "■■"
            if not (ndcg_match and hit_match):
                all_match = False

            print("{:<20} {:.4f} ± {:.4f}        {:.4f} ± {:.4f}        {}".format(
                epsilon, curr_ndcg, curr["NDCG@10"]["std"],
                curr_hit, curr["Hit@10"]["std"], status))
```

```
# Section 2: Heterogeneity Sweep Results (RQ3)
print("\n" + "=" * 80)
print("SECTION 2: HETEROGENEITY SWEEP RESULTS (RQ3: Data Distribution Impact)")
print("=" * 80)
print("\nConfiguration: 100 clients, 10 rounds, no DP")
print("\n{:<20} {:<25} {:<25} {:<10}".format(
    "Alpha (heterogeneity)", "NDCG@10", "Hit@10", "Status"))
print("-" * 80)

hetero_configs = [
    ("0.1", "dp_inf_alpha_0.1_dim_64_clients_100"),
    ("0.5", "dp_inf_alpha_0.5_dim_64_clients_100"),
    ("1.0", "dp_inf_alpha_1.0_dim_64_clients_100"),
]

for alpha, config_key in hetero_configs:
    if config_key in current_results and config_key in published:
        curr = current_results[config_key]
        pub = published[config_key]

        curr_ndcg = curr["NDCG@10"]["mean"]
        pub_ndcg = pub["NDCG@10"]["mean"]
        curr_hit = curr["Hit@10"]["mean"]
        pub_hit = pub["Hit@10"]["mean"]

        ndcg_match = abs(curr_ndcg - pub_ndcg) < 0.0001
        hit_match = abs(curr_hit - pub_hit) < 0.0001

        status = "■" if (ndcg_match and hit_match) else "■■"
        if not (ndcg_match and hit_match):
            all_match = False

        print("{:<20} {:.4f} ± {:.4f}        {:.4f} ± {:.4f}        {}".format(
            alpha, curr_ndcg, curr["NDCG@10"]["std"],
            curr_hit, curr["Hit@10"]["std"], status))

# Section 3: Centralized Baseline
print("\n" + "=" * 80)
print("SECTION 3: CENTRALIZED BASELINE")
print("=" * 80)

if "centralized" in current_results:
    curr = current_results["centralized"]
    pub = published["centralized"]

    print(f"\nNDCG@10: {curr['NDCG@10']:.4f} (Published: {pub['NDCG@10']:.4f})")
    print(f"Hit@10:  {curr['Hit@10']:.4f} (Published: {pub['Hit@10']:.4f})")

    ndcg_match = abs(curr['NDCG@10'] - pub['NDCG@10']) < 0.0001
    hit_match = abs(curr['Hit@10'] - pub['Hit@10']) < 0.0001

    if ndcg_match and hit_match:
        print("\nStatus: ■ Matches published baseline")
    else:
        print("\nStatus: ■■ Differs from published baseline")
        all_match = False

# Section 4: Attack Evaluation
print("\n" + "=" * 80)
print("SECTION 4: PRIVACY ATTACK EVALUATION (RQ2)")
print("=" * 80)

attack_file = Path("results/attack_evaluation_summary.json")
if attack_file.exists():
    with open(attack_file) as f:
        attacks = json.load(f)

    print("\n{:<15} {:<12} {:<15} {:<20}".format(
        "DP Budget (ε)", "MIA AUC", "MIA Accuracy", "Inversion Top-K"))
    print("-" * 80)

    for eps in ["inf", "8", "4", "2", "1"]:
        if eps in attacks:
```

```
                mia = attacks[eps].get("mia", {})
                inv = attacks[eps].get("inversion", {})
                print("{:<15} {:.4f}       {:.4f}          {:.4f}".format(
                    eps, mia.get("auc", 0), mia.get("accuracy", 0),
                    inv.get("top_k_accuracy", 0)))
    else:
        print("\n■■  Attack evaluation results not found")

    # Final Summary
    print("\n" + "=" * 80)
    print("VERIFICATION SUMMARY")
    print("=" * 80)

    print(f"\nTotal configurations verified: {len(published)}")
    print(f"Configurations found: {len(current_results)}")

    if all_match:
        print("\n■ ALL RESULTS MATCH PUBLISHED VALUES!")
        print("\nConclusion: The experiments are reproducible and the results")
        print("in the results/ directory match exactly with those published")
        print("in EXPERIMENT_STATUS.md")
    else:
        print("\n■■  Some results differ from published values")
        print("\nThis could be due to:")
        print("  - Results from a different run")
        print("  - Updated experiment configuration")
        print("  - Random seed variation")

    print("\n" + "=" * 80)
    print("RECOMMENDATION")
    print("=" * 80)

    if all_match:
        print("\nThe existing results are valid and match published values.")
        print("No need to re-run experiments unless:")
        print("  1. You want to verify reproducibility with different seeds")
        print("  2. You want to test with different configurations")
        print("  3. You want to update the published results")
        print("\nTo re-run all experiments:")
        print("  python run_complete_experiment.py")
        print("\nEstimated time: ~60 minutes")
    else:
        print("\nConsider re-running experiments to verify reproducibility:")
        print("  python run_complete_experiment.py")

    print("\n" + "=" * 80)


if __name__ == "__main__":
    generate_report()
```

File: federated_learning_in_mobile/EASY_RESULTS_ACCESS.md

```
# Easy Results Access Guide

## ■ Results Now Saved to Downloads Folder!

The app now saves all results to an **easily accessible location**:

### ■ Location
``
/storage/emulated/0/Download/FL_Results/
``

This is the standard **Downloads folder** on Android, which you can access using any file manager app!

---

## ■ How to Access Files (3 Easy Ways)

### Method 1: Android File Manager (Easiest!)

1. **Open File Manager** on your Android device
```

2. **Tap "Downloads"** (or "Internal Storage" → "Download")
3. **Look for "FL_Results" folder**
4. **Open the folder** - you'll see:
   - `dp_inf_..._device_XXX_t1234567890.json` (full results)
   - `dp_inf_..._device_XXX_t1234567890_summary.csv` (summary table)

That's it! You can now:
- Copy files to your PC via USB
- Share files via email/WhatsApp/Drive
- Open and view files directly

Method 2: Share Button (Built-in!)

1. In the app, click **"Show Results Location"** button
2. Click **"Share Files"** button
3. Choose where to share:
   - Email
   - Google Drive
   - WhatsApp
   - Bluetooth
   - Any app that accepts files

Super easy! The app will share the latest JSON file automatically.

Method 3: ADB (For Developers)

If you have ADB set up:

```bash
# Pull all results
adb pull /storage/emulated/0/Download/FL_Results/ ./mobile_results/

# Or pull specific file
adb pull /storage/emulated/0/Download/FL_Results/dp_inf_*.json
```

■ What Files Are Saved?

JSON File (`experiment_id.json`)
- **Full detailed data** for analysis
- All training rounds
- Metrics, resource usage, timestamps
- Same format as Python experiments

CSV File (`experiment_id_summary.csv`)
- **Quick summary table**
- One row per training round
- Easy to open in Excel/Google Sheets
- Perfect for quick analysis

■ Finding Your Files

File Naming
Files are named with the experiment ID:
```
dp_inf_alpha_null_dim_16_clients_1_device_Samsung_Galaxy_t1701234567890.json
```

The name includes:
- DP epsilon (dp_inf = no privacy)
- Embedding dimension
- Device ID
- Timestamp

Viewing Files
- **JSON**: Open with any text editor or JSON viewer

```
- **CSV**: Open with Excel, Google Sheets, or any spreadsheet app

---

## ■ New Features

1. ■ **Downloads Folder** - Easy access via file manager
2. ■ **Share Button** - Share files directly from app
3. ■ **File List** - See all saved files in the app
4. ■ **Better Path Display** - Shows exact location

---

## ■ Quick Start

1. **Run training rounds** in the app
2. **After each round**, files are automatically saved
3. **Open File Manager** → Downloads → FL_Results
4. **Copy files** to your PC for analysis!

---

## ■ Example Workflow

1. Connect app to server ■
2. Run 3 training rounds ■
3. Open File Manager ■
4. Go to Downloads/FL_Results ■
5. Copy both JSON and CSV files ■
6. Transfer to PC (via USB/email/Drive) ■
7. Analyze in Python/Excel ■

---

## ■■ Troubleshooting

**Q: Can't find FL_Results folder?**
- Make sure you've run at least one training round
- Check if Downloads folder exists
- Try using "Share Files" button instead

**Q: Files not appearing?**
- Check app logs for errors
- Verify app has storage permissions
- Try running training round again

**Q: Want to change location?**
- Currently hardcoded to Downloads folder
- Most accessible location for users
- Can be modified in `metrics_collector.dart` if needed

---

## ■ For Your Thesis

Now you can easily:
1. ■ Collect results from mobile devices
2. ■ Access files without ADB
3. ■ Share results between devices
4. ■ Combine with Python experiment results
5. ■ Analyze everything together

**Perfect for thesis data collection!** ■
```

File: federated_learning_in_mobile/FIX_PATH_PROVIDER_ERROR.md

```
# Fix for "Unable to establish connection on channel: path_provider" Error

## What I Fixed

1. ■ Added storage permissions to AndroidManifest.xml
2. ■ Made metrics collector initialization non-blocking (won't prevent connection)
3. ■ Added fallback storage options (external storage, temporary directory)
```

````
4. ■ Better error handling

## How to Apply the Fix

### Step 1: Stop the app completely
- Close the app on your device
- Or press `q` in the terminal where `flutter run` is running

### Step 2: Rebuild the app (IMPORTANT!)
You MUST do a full rebuild, not just hot reload:

```powershell
cd C:\Users\jonat\PycharmProjects\master_thesis\federated_learning_in_mobile

# Clean build (optional but recommended)
flutter clean

# Get dependencies
flutter pub get

# Rebuild and run
flutter run
```

**DO NOT use hot reload** (`r` key) - this won't fix the plugin issue!

### Step 3: Try connecting again

The connection should work now. Even if metrics collection fails, the connection will still work.

## What Changed

### Before:
- Metrics collector failure blocked connection ■
- No fallback storage options ■

### After:
- Connection works even if metrics fail ■
- Multiple storage fallback options ■
- Better error messages ■

## If It Still Doesn't Work

### Option 1: Disable metrics temporarily
If you just want to test the connection, you can comment out metrics initialization in `lib/main.dart`:

```dart
// Temporarily disable metrics to test connection
// _metricsCollector = MetricsCollector(experimentId: experimentId);
// await _metricsCollector!.initialize();
```

### Option 2: Check device/emulator
Make sure you're using a real device or properly configured emulator:

```powershell
flutter devices
```

### Option 3: Uninstall and reinstall
Sometimes old app data causes issues:

```powershell
# Uninstall the app
adb uninstall com.example.federated_learning_in_mobile

# Reinstall
flutter run
```

## Expected Behavior After Fix

1. ■ App launches without errors
2. ■ "Connect to Server" works
````

```
3. ■ Connection succeeds even if metrics initialization has issues
4. ■ You'll see warnings in logs, but connection continues
5. ■ Training rounds work normally

## Testing

After rebuilding, try:
1. Connect to server - should work ■
2. Run training round - should work ■
3. Check logs for metrics status:
   - ■ Good: "Metrics collector initialized"
   - ■■ Warning: "Metrics collector initialization failed" (but connection still works)
   - ■ Error: Connection blocked (then we need more fixes)

Let me know what happens after the full rebuild!
```

File: federated_learning_in_mobile/METRICS_COLLECTION.md

```
# Metrics Collection Guide

The mobile app now automatically collects and saves training metrics in the same format as the Python exper
iments.

## How It Works

1. **Automatic Collection**: When you connect to the server and run training rounds, metrics are automatica
lly collected.

2. **Saved After Each Round**: After each training round completes, results are saved to:
   - JSON file (full detailed data)
   - CSV file (summary for easy analysis)

3. **File Location**: Results are saved in the app's documents directory:
   ```
   /data/data/com.example.federated_learning_in_mobile/app_flutter/fl_results/
   ```

## What Metrics Are Collected

### Per Training Round:
- Training loss
- Number of samples
- Training time (ms)
- Resource metrics:
  - Battery level and drain
  - Memory usage
  - CPU usage
  - Network bandwidth

### Experiment Configuration:
- Number of users and items
- Embedding dimension
- Client ID
- Device ID
- Server URL

## File Names

Files are named using the experiment ID format:
```
dp_inf_alpha_null_dim_16_clients_1_device_XXX_t1234567890.json
dp_inf_alpha_null_dim_16_clients_1_device_XXX_t1234567890_summary.csv
```

The experiment ID includes:
- DP epsilon (dp_inf = no privacy)
- Alpha (heterogeneity parameter)
- Embedding dimension
- Number of clients
- Device ID
- Timestamp

## Accessing Results
```

````
### Option 1: Using ADB (Recommended)
```bash
# Pull all results from device
adb pull /data/data/com.example.federated_learning_in_mobile/app_flutter/fl_results/ ./mobile_results/

# Or pull specific file
adb pull /data/data/com.example.federated_learning_in_mobile/app_flutter/fl_results/dp_inf_*.json
```

### Option 2: Using Android File Manager
1. Connect device via USB
2. Enable file transfer mode
3. Navigate to: Android/data/com.example.federated_learning_in_mobile/files/fl_results/
4. Copy files to your computer

### Option 3: Using the App
1. After running at least one training round
2. Click "Show Results Location" button
3. Copy the file path shown

## File Formats

### JSON Format
The JSON file contains all collected data in a structured format:
```json
{
  "experiment_id": "...",
  "timestamp": "2025-11-29T...",
  "config": {
    "num_users": 50,
    "num_items": 4032,
    ...
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 0.45,
      "resource_metrics": {
        "battery_drain": 2.5,
        "training_time_ms": 150,
        ...
      },
      ...
    }
  ],
  "client_metrics": [...],
  "mobile_metrics": [...]
}
```

### CSV Format
The CSV file is a summary table with key metrics per round:
- Round number
- Train Loss
- Training Time (MS)
- Battery Drain
- Number of Samples
- And other metrics

## Using Results for Thesis

The mobile results can be:
1. **Combined with Python results** - Same format, easy to merge
2. **Analyzed separately** - Mobile-specific resource metrics
3. **Compared** - Mobile vs Python client performance

### Analysis Example
```python
import json
import pandas as pd

# Load mobile results
with open('mobile_results/dp_inf_..._device_XXX.json') as f:
````
````

````
    mobile_data = json.load(f)

# Extract metrics
rounds = mobile_data['rounds']
battery_drain = [r['resource_metrics']['battery_drain'] for r in rounds]
training_times = [r['resource_metrics']['training_time_ms'] for r in rounds]

# Plot resource usage
import matplotlib.pyplot as plt
plt.plot(battery_drain, label='Battery Drain (%)')
plt.plot(training_times, label='Training Time (ms)')
plt.legend()
plt.show()
```

## Troubleshooting

**Q: Files not appearing?**
- Make sure you've run at least one training round
- Check app logs for errors
- Verify app has storage permissions

**Q: Can't access files via ADB?**
- Make sure USB debugging is enabled
- Try: `adb shell run-as com.example.federated_learning_in_mobile ls app_flutter/fl_results/`

**Q: Want to reset/clear old results?**
- Uninstall and reinstall the app (clears app data)
- Or manually delete files via ADB

## Next Steps

1. Run training rounds from mobile app
2. Collect results using ADB
3. Combine with Python experiment results
4. Analyze for your thesis!
````

File: federated_learning_in_mobile/README.md

````
# Federated Learning Mobile Client

A Flutter-based Android application for participating in federated learning experiments for movie recommend
ation systems.

## Overview

This mobile client allows Android devices to participate as federated learning clients in a distributed tra
ining system. It implements:

- Matrix Factorization model in pure Dart
- Federated learning client logic
- Resource monitoring (battery, CPU, memory)
- Integration with Python FastAPI server

## Features

- ■ Connect to federated learning server
- ■ Download global model parameters
- ■ Train locally on device data
- ■ Upload model updates to server
- ■ Resource usage monitoring
- ■ Real-time training metrics display

## Prerequisites

1. **Flutter SDK** (3.9.2 or higher)
   ```bash
   flutter --version
   ```

2. **Android Studio** or Android SDK for development

3. **Python Server** running (see parent directory `server.py`)
````
````

   - Server should be accessible from your device
   - Use ngrok or local network IP for mobile access

Setup Instructions

1. Install Dependencies

```bash
cd federated_learning_in_mobile
flutter pub get
```

2. Configure Server URL

1. Find your server's IP address:
   - **For local network**: Use your PC's local IP (e.g., `192.168.1.100:8000`)
   - **For ngrok**: Use the ngrok URL (e.g., `https://abc123.ngrok.io`)

2. Update the server URL in the app when you launch it, or modify the default in `lib/main.dart`:

```dart
text: 'http://YOUR_IP:8000', // Replace with your server address
```

3. Build and Run

On Android Device:
```bash
# Connect your Android device via USB with USB debugging enabled
flutter devices  # Check if device is detected
flutter run
```

On Android Emulator:
```bash
# Start an Android emulator from Android Studio
flutter emulators  # List available emulators
flutter run
```

Usage

1. Start the Python Server

First, start your federated learning server:

```bash
# In the parent directory
python server.py
```

Or use the helper script:
```bash
python start_server.py
```

Important: Make sure the server is accessible from your mobile device:
- If using local network: Ensure both devices are on the same WiFi network
- Use your PC's local IP address (not `localhost`)
- Example: `http://192.168.1.100:8000`

2. Launch the Mobile App

1. Open the app on your Android device/emulator
2. Enter your server URL (e.g., `http://192.168.1.100:8000`)
3. Enter a unique Client ID (default is auto-generated)
4. Click "Connect to Server"

3. Run Training Rounds

Once connected:
1. Click "Run Training Round"
2. The app will:
   - Fetch the global model from server

````
   - Train locally on sample data
   - Upload updated parameters
   - Display metrics and resource usage

## Architecture

```
lib/
■■■ main.dart                    # Main UI and app entry point
■■■ models/
■   ■■■ matrix_factorization.dart  # Matrix Factorization model (pure Dart)
■■■ services/
■   ■■■ api_client.dart         # HTTP client for server communication
■   ■■■ fl_client.dart          # Federated learning client logic
■■■ utils/
    ■■■ model_encoder.dart      # Model parameter encoding/decoding
    ■■■ resource_monitor.dart   # Resource usage monitoring
```

## Integration with Python Server

The mobile client communicates with the Python FastAPI server using JSON-based parameter transfer:

- **Model Parameters**: Base64-encoded Float32 arrays
- **Endpoints**: `/register`, `/global-params-json`, `/upload-params-json`
- **Wire Format**: Compatible with Python server's portable JSON format

## For Thesis Experiments

### Running Experiments with Multiple Devices

1. **Device 1**: Launch app, connect to server, set Client ID to `android_device_1`
2. **Device 2**: Launch app, connect to server, set Client ID to `android_device_2`
3. **Server**: Initialize model and run aggregation rounds

### Data Collection

The app collects and displays:
- Training metrics (loss, samples, training time)
- Resource metrics (battery level, battery drain, device info)
- Network usage (implicit in upload/download)

For detailed logging, check the app's log output or export metrics.

### Combining with Python Clients

You can run hybrid experiments:
- Multiple Python clients (for large-scale sweeps)
- 2+ Android devices (for real mobile validation)
- All connect to the same server
- Server aggregates all updates together

## Troubleshooting

### Connection Issues

**Problem**: Cannot connect to server
- **Solution**:
  - Ensure server is running
  - Check server URL is correct (use IP, not localhost)
  - Verify both devices are on same network
  - Check firewall settings

**Problem**: Connection timeout
- **Solution**:
  - Use ngrok for external access
  - Check server logs for errors
  - Verify server port (default: 8000)

### Model Loading Issues

**Problem**: Model parameters not loading correctly
- **Solution**:
  - Verify Float32 encoding compatibility
````

```
  - Check server model dimensions match client expectations
  - Review model_encoder.dart for byte order (endianness)

### Resource Monitoring

**Problem**: Battery/memory metrics not accurate
- **Solution**:
  - Resource monitoring uses platform APIs
  - For detailed profiling, use Android Profiler
  - CPU/memory monitoring may require additional permissions

## Development Notes

### Model Compatibility

The Dart Matrix Factorization model is designed to be compatible with the Python PyTorch version:
- Same embedding dimensions
- Same initialization scheme
- Same forward pass logic
- Parameter format matches server expectations

### Performance Considerations

For mobile devices:
- Model size should be small (embedding_dim ≤ 32 recommended)
- Local training epochs should be minimal (1-2 epochs)
- Batch size should be small (16-32)
- Use sample data sets for mobile devices (5-50 samples)

## Future Enhancements

- [ ] Load actual MovieLens data from device storage
- [ ] Implement DP-SGD for privacy-preserving training
- [ ] Add background training capability
- [ ] Export metrics to CSV/JSON for analysis
- [ ] Add visualization of training progress
- [ ] Implement model checkpointing

## Thesis Integration

This mobile client addresses the requirement for **physical device participation**:

1. **Real Device Validation**: Test actual mobile constraints (battery, memory, network)
2. **Resource Profiling**: Collect real-world resource usage metrics
3. **Mobile Deployment**: Demonstrate feasibility on actual Android devices
4. **Hybrid Experiments**: Combine Python simulation with real mobile devices

## License

This project is part of a Master's thesis research project.

## Contact

For questions or issues, refer to the main thesis repository.
```

File: federated_learning_in_mobile/README_DEMO_RECOMMENDATIONS.md

---

```
# Demo Recommendations Flow

This mobile app now includes a presentation-ready recommendation demo:

- Dashboard remains unchanged for training metrics and experiment controls.
- A separate `Movie Recommendations` screen shows ranked recommendations.
- `Demo Mode: Train and Show Recommendations` runs one round and navigates automatically.

## Data Source for Titles

Movie names are loaded from:

- `assets/movielens_titles.csv`

This file maps model `item_id` (0-based) to MovieLens titles.

## Quick Demo Steps
```

````
1. Connect to server in the dashboard.
2. Press `Demo Mode: Train and Show Recommendations`.
3. Show:
   - Dashboard logs and metrics
   - Recommendations screen with title, score explanation, seen/unseen status

## Rebuild title asset (optional)

Helper script:

- `tooling/build_movie_titles_asset.py`
````

File: federated_learning_in_mobile/SETUP_COMPLETE.md

````
# ■ Federated Learning Mobile Client - Setup Complete!

## What Has Been Built

### ■ Complete Flutter Application

1. **Matrix Factorization Model** (`lib/models/matrix_factorization.dart`)
   - Pure Dart implementation
   - Compatible with Python server
   - Supports training and prediction

2. **API Client** (`lib/services/api_client.dart`)
   - HTTP communication with FastAPI server
   - Register, fetch model, upload parameters
   - Error handling

3. **Federated Learning Client** (`lib/services/fl_client.dart`)
   - Complete FL workflow
   - Local training
   - Model synchronization

4. **Resource Monitoring** (`lib/utils/resource_monitor.dart`)
   - Battery level tracking
   - Device information
   - Resource metrics collection

5. **Model Encoder** (`lib/utils/model_encoder.dart`)
   - Base64 encoding/decoding
   - Float32 parameter encoding
   - Compatible with Python server format

6. **Mobile UI** (`lib/main.dart`)
   - Server configuration
   - Training controls
   - Real-time metrics display
   - Logs viewer

7. **Android Permissions**
   - Network access configured
   - Ready for deployment

## Next Steps

### 1. Install Dependencies

```bash
cd federated_learning_in_mobile
flutter pub get
```

### 2. Test Build

```bash
# Check Flutter setup
flutter doctor

# Build APK (to verify no errors)
flutter build apk --debug
````
````

````markdown
```

### 3. Run on Device/Emulator

```bash
# List available devices
flutter devices

# Run on connected device
flutter run

# Or run on specific device
flutter run -d <device_id>
```

### 4. Connect to Server

1. **Find your server URL:**
   - Local network: Your PC's IP address (e.g., `192.168.1.100:8000`)
   - ngrok: Your ngrok URL

2. **In the app:**
   - Enter server URL
   - Enter unique client ID (e.g., `android_device_1`)
   - Click "Connect to Server"

3. **Verify connection:**
   - Status should show "Connected"
   - Logs should show registration success

### 5. Run Training Round

1. Click "Run Training Round"
2. Watch logs for progress
3. Check metrics after completion

## Testing Checklist

- [ ] App builds without errors
- [ ] App launches on device/emulator
- [ ] Can connect to server (local network or ngrok)
- [ ] Model parameters download correctly
- [ ] Training runs successfully
- [ ] Parameters upload to server
- [ ] Resource metrics are displayed

## Troubleshooting

**Build Errors:**
```bash
flutter clean
flutter pub get
flutter pub upgrade
```

**Connection Issues:**
- Use your PC's IP address (not `localhost`)
- Ensure server is running
- Check firewall settings
- Try ngrok as alternative

**Model Loading Issues:**
- Verify server model dimensions
- Check server logs
- Ensure Float32 encoding is compatible

## For Thesis Experiments

See `EXPERIMENT_GUIDE.md` in the parent directory for:
- How to run Python + Android experiments together
- Data collection strategies
- Analysis workflow
- Thesis integration guidance
````
````

```
## File Structure

```
federated_learning_in_mobile/
■■■ lib/
■   ■■■ main.dart                     # Main UI
■   ■■■ models/
■   ■   ■■■ matrix_factorization.dart  # MF model
■   ■■■ services/
■   ■   ■■■ api_client.dart           # Server communication
■   ■   ■■■ fl_client.dart            # FL logic
■   ■■■ utils/
■       ■■■ model_encoder.dart        # Parameter encoding
■       ■■■ resource_monitor.dart     # Resource tracking
■■■ android/                          # Android configuration
■■■ pubspec.yaml                      # Dependencies
■■■ README.md                         # Documentation
```

Ready for Thesis! ■

You now have:
- ■ Working Android federated learning client
- ■ Integration with Python server
- ■ Resource monitoring
- ■ Complete documentation
- ■ Experiment guide

You're ready to start collecting data for your thesis!
```

File: federated_learning_in_mobile/UI_REDESIGN_SUMMARY.md

```
# ■ Federated Learning Mobile App - UI Redesign

## Overview
Successfully redesigned the mobile app dashboard to match the target design with a modern, dark theme inter
face.

## Changes Made

### ■ Theme & Colors
- **Dark Navy Background**: `#1A2332` (main background)
- **Card Surface Color**: `#212D3F` (elevated surfaces)
- **Cyan Accent**: `#00BCD4` (primary actions, highlights)
- **Green Status**: `#4CAF50` (status indicators)
- **Dark Console**: `#0D1117` (log terminal background)

### ■ Layout Changes

#### 1. **Header Section**
- ■ Removed AppBar
- ■ Added custom header with "Federated Client" title
- ■ Added "Researcher Edition v2.4" subtitle
- ■ Added settings icon button in top-right corner
- ■ Modern spacing and typography

#### 2. **Server Configuration Card**
- ■ Section header with DNS icon
- ■ Uppercase "SERVER CONFIGURATION" label
- ■ Cyan-highlighted "SERVER URL" label
- ■ Grey "CLIENT ID" label
- ■ Dark input fields with proper styling
- ■ Large cyan "CONNECT TO SERVER" button with lightning icon
- ■ Proper disabled states

#### 3. **Status Section Card**
- ■ Green dot status indicator
- ■ "Status: Ready" text
- ■ "IDLE" / "TRAINING" badge in top-right
- ■ Two large square action buttons side-by-side:
  - "Run 1 Round" with play icon
  - "Run 10 Rounds" with fast-forward icon
- ■ Cyan icons on dark button backgrounds
```

- ■ Upload and Location buttons (when training is complete)

4. **Live Logs Section**
- ■ "LIVE LOGS" header with TV icon
- ■ "CLEAR CONSOLE" button
- ■ Dark terminal-style log display
- ■ Color-coded log messages:
  - ■ **SYSTEM:** Green (#4CAF50)
  - ■ **NETWORK:** Cyan (#00BCD4)
  - ■ **INFO:** Blue (#58A6FF)
  - ■ **WARN:** Orange (#FFA726)
  - ■ **ERROR:** Red (#EF5350)
- ■ Courier font for logs
- ■ Empty state message
- ■ 250px height scrollable container

■■ Removed Sections
- ■ Removed "Training Metrics" card (consolidated into logs)
- ■ Removed "Resource Metrics" card (consolidated into logs)
- ■ Cleaned up unused code (_lastMetrics, _resourceMetrics, _buildMetricRow)

■ Log Messages Enhanced
- ■ Added proper prefixes: SYSTEM:, NETWORK:, INFO:, WARN:, ERROR:
- ■ Updated connection logs:
  - "NETWORK: DNS lookup resolved for coordinator."
  - "INFO: Handshake protocol version 3.2.0."
  - "SYSTEM: Client initialized successfully."
  - "SYSTEM: Local model weight buffer allocated (512MB)."
  - "WARN: Model not initialized on server yet."

■ Technical Improvements
- ■ Fixed all compilation errors
- ■ Removed deprecated color properties (background → surface)
- ■ Fixed CardTheme type (CardTheme → CardThemeData)
- ■ Fixed withOpacity() deprecation (→ withValues())
- ■ Added SafeArea for proper layout
- ■ Improved code structure and readability

Result
The app now has a professional, modern UI that matches the target design:
- Dark theme with proper color scheme
- Clean, organized layout
- Better visual hierarchy
- Enhanced user experience
- Production-ready appearance

Testing
To see the changes:
```bash
cd federated_learning_in_mobile
flutter run
```

Screenshot Comparison
Before: Light theme with AppBar, multiple metric cards
After: Dark navy theme, streamlined layout, modern design matching target

---
Status: ■ Complete - All changes implemented and tested
Date: 2026-03-03

File: federated_learning_in_mobile/.dart_tool/package_config.json

---


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    {
```

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      "packageUri": "lib/",
      "languageVersion": "3.1"
    },
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      "name": "sync_http",
      "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/sync_http-0.3.1",
      "packageUri": "lib/",
      "languageVersion": "2.12"
    },
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      "name": "term_glyph",
      "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/term_glyph-1.2.2",
      "packageUri": "lib/",
      "languageVersion": "3.1"
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      "packageUri": "lib/",
      "languageVersion": "3.7"
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      "packageUri": "lib/",
      "languageVersion": "3.5"
    },
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      "name": "upower",
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      "languageVersion": "2.14"
    },
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      "packageUri": "lib/",
      "languageVersion": "3.8"
    },
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      "name": "url_launcher_platform_interface",
      "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/url_launcher_platform_interface-2.3.2",
      "packageUri": "lib/",
      "languageVersion": "3.1"
    },
    {
      "name": "url_launcher_web",
      "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/url_launcher_web-2.4.1",
      "packageUri": "lib/",
      "languageVersion": "3.6"
```

```
    },
    {
      "name": "url_launcher_windows",
      "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/url_launcher_windows-3.1.5",
      "packageUri": "lib/",
      "languageVersion": "3.8"
    },
    {
      "name": "uuid",
      "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/uuid-4.5.2",
      "packageUri": "lib/",
      "languageVersion": "3.0"
    },
    {
      "name": "vector_math",
      "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/vector_math-2.2.0",
      "packageUri": "lib/",
      "languageVersion": "3.1"
    },
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      "name": "vm_service",
      "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/vm_service-15.0.2",
      "packageUri": "lib/",
      "languageVersion": "3.5"
    },
    {
      "name": "web",
      "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/web-1.1.1",
      "packageUri": "lib/",
      "languageVersion": "3.4"
    },
    {
      "name": "webdriver",
      "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/webdriver-3.1.0",
      "packageUri": "lib/",
      "languageVersion": "3.1"
    },
    {
      "name": "win32",
      "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/win32-5.15.0",
      "packageUri": "lib/",
      "languageVersion": "3.8"
    },
    {
      "name": "win32_registry",
      "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/win32_registry-1.1.5",
      "packageUri": "lib/",
      "languageVersion": "3.4"
    },
    {
      "name": "xdg_directories",
      "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/xdg_directories-1.1.0",
      "packageUri": "lib/",
      "languageVersion": "3.3"
    },
    {
      "name": "xml",
      "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/xml-6.6.1",
      "packageUri": "lib/",
      "languageVersion": "3.8"
    },
    {
      "name": "federated_learning_in_mobile",
      "rootUri": "../",
      "packageUri": "lib/",
      "languageVersion": "3.9"
    }
  ],
  "generator": "pub",
  "generatorVersion": "3.11.3",
  "flutterRoot": "file:///opt/homebrew/share/flutter",
  "flutterVersion": "3.41.5",
  "pubCache": "file:///Users/jonat/.pub-cache"
}
```

File: federated_learning_in_mobile/.dart_tool/package_graph.json

----


```
{
  "roots": [
    "federated_learning_in_mobile"
  ],
  "packages": [
    {
      "name": "federated_learning_in_mobile",
      "version": "1.0.0+1",
      "dependencies": [
        "async",
        "battery_plus",
        "csv",
        "cupertino_icons",
        "device_info_plus",
        "flutter",
        "http",
        "open_filex",
        "path_provider",
        "share_plus"
      ],
      "devDependencies": [
        "flutter_lints",
        "flutter_test",
        "integration_test"
      ]
    },
    {
      "name": "flutter_lints",
      "version": "5.0.0",
      "dependencies": [
        "lints"
      ]
    },
    {
      "name": "integration_test",
      "version": "0.0.0",
      "dependencies": [
        "flutter",
        "flutter_driver",
        "flutter_test",
        "path",
        "vm_service"
      ]
    },
    {
      "name": "flutter_test",
      "version": "0.0.0",
      "dependencies": [
        "clock",
        "collection",
        "fake_async",
        "flutter",
        "leak_tracker_flutter_testing",
        "matcher",
        "meta",
        "path",
        "stack_trace",
        "stream_channel",
        "test_api",
        "vector_math"
      ]
    },
    {
      "name": "open_filex",
      "version": "4.7.0",
      "dependencies": [
        "ffi",
        "flutter"
      ]
    },
    {
      "name": "share_plus",
```

```
      "version": "7.2.2",
      "dependencies": [
        "cross_file",
        "ffi",
        "file",
        "flutter",
        "flutter_web_plugins",
        "meta",
        "mime",
        "share_plus_platform_interface",
        "url_launcher_linux",
        "url_launcher_platform_interface",
        "url_launcher_web",
        "url_launcher_windows",
        "win32"
      ]
    },
    {
      "name": "csv",
      "version": "5.1.1",
      "dependencies": []
    },
    {
      "name": "path_provider",
      "version": "2.1.5",
      "dependencies": [
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        "path_provider_android",
        "path_provider_foundation",
        "path_provider_linux",
        "path_provider_platform_interface",
        "path_provider_windows"
      ]
    },
    {
      "name": "async",
      "version": "2.13.0",
      "dependencies": [
        "collection",
        "meta"
      ]
    },
    {
      "name": "battery_plus",
      "version": "6.2.3",
      "dependencies": [
        "battery_plus_platform_interface",
        "flutter",
        "flutter_web_plugins",
        "meta",
        "upower",
        "web"
      ]
    },
    {
      "name": "device_info_plus",
      "version": "10.1.2",
      "dependencies": [
        "device_info_plus_platform_interface",
        "ffi",
        "file",
        "flutter",
        "flutter_web_plugins",
        "meta",
        "web",
        "win32",
        "win32_registry"
      ]
    },
    {
      "name": "http",
      "version": "1.6.0",
      "dependencies": [
        "async",
```

```
        "http_parser",
        "meta",
        "web"
      ]
    },
    {
      "name": "cupertino_icons",
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      "dependencies": []
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    {
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      "version": "0.0.0",
      "dependencies": [
        "characters",
        "collection",
        "material_color_utilities",
        "meta",
        "sky_engine",
        "vector_math"
      ]
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    {
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    {
      "name": "vm_service",
      "version": "15.0.2",
      "dependencies": []
    },
    {
      "name": "path",
      "version": "1.9.1",
      "dependencies": []
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      "version": "0.0.0",
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        "flutter",
        "flutter_test",
        "fuchsia_remote_debug_protocol",
        "matcher",
        "meta",
        "path",
        "vm_service",
        "webdriver"
      ]
    },
    {
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      ]
    },
    {
      "name": "meta",
      "version": "1.17.0",
      "dependencies": []
    },
    {
      "name": "collection",
      "version": "1.19.1",
      "dependencies": []
    },
    {
      "name": "leak_tracker_flutter_testing",
      "version": "3.0.10",
      "dependencies": [
        "flutter",
```

```
        "leak_tracker",
        "leak_tracker_testing",
        "matcher",
        "meta"
      ]
    },
    {
      "name": "vector_math",
      "version": "2.2.0",
      "dependencies": []
    },
    {
      "name": "stack_trace",
      "version": "1.12.1",
      "dependencies": [
        "path"
      ]
    },
    {
      "name": "clock",
      "version": "1.1.2",
      "dependencies": []
    },
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        "collection"
      ]
    },
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        "meta",
        "stack_trace",
        "term_glyph",
        "test_api"
      ]
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      "dependencies": [
        "async",
        "boolean_selector",
        "collection",
        "meta",
        "source_span",
        "stack_trace",
        "stream_channel",
        "string_scanner",
        "term_glyph"
      ]
    },
    {
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      "dependencies": []
    },
    {
      "name": "win32",
      "version": "5.15.0",
      "dependencies": [
        "ffi"
      ]
    },
    {
      "name": "url_launcher_platform_interface",
      "version": "2.3.2",
      "dependencies": [
        "flutter",
```

```
        "plugin_platform_interface"
      ]
    },
    {
      "name": "url_launcher_linux",
      "version": "3.2.2",
      "dependencies": [
        "flutter",
        "url_launcher_platform_interface"
      ]
    },
    {
      "name": "url_launcher_windows",
      "version": "3.1.5",
      "dependencies": [
        "flutter",
        "url_launcher_platform_interface"
      ]
    },
    {
      "name": "url_launcher_web",
      "version": "2.4.1",
      "dependencies": [
        "flutter",
        "flutter_web_plugins",
        "url_launcher_platform_interface",
        "web"
      ]
    },
    {
      "name": "file",
      "version": "7.0.1",
      "dependencies": [
        "meta",
        "path"
      ]
    },
    {
      "name": "share_plus_platform_interface",
      "version": "3.4.0",
      "dependencies": [
        "cross_file",
        "flutter",
        "meta",
        "mime",
        "path_provider",
        "plugin_platform_interface",
        "uuid"
      ]
    },
    {
      "name": "flutter_web_plugins",
      "version": "0.0.0",
      "dependencies": [
        "flutter"
      ]
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    {
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      "version": "0.3.5+1",
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        "meta",
        "web"
      ]
    },
    {
      "name": "path_provider_windows",
      "version": "2.3.0",
      "dependencies": [
```

```
        "ffi",
        "flutter",
        "path",
        "path_provider_platform_interface"
      ]
    },
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      "version": "2.1.2",
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        "platform",
        "plugin_platform_interface"
      ]
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        "ffi",
        "flutter",
        "path",
        "path_provider_platform_interface",
        "xdg_directories"
      ]
    },
    {
      "name": "path_provider_foundation",
      "version": "2.5.1",
      "dependencies": [
        "flutter",
        "path_provider_platform_interface"
      ]
    },
    {
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      "version": "2.2.22",
      "dependencies": [
        "flutter",
        "path_provider_platform_interface"
      ]
    },
    {
      "name": "web",
      "version": "1.1.1",
      "dependencies": []
    },
    {
      "name": "upower",
      "version": "0.7.0",
      "dependencies": [
        "dbus"
      ]
    },
    {
      "name": "battery_plus_platform_interface",
      "version": "2.0.1",
      "dependencies": [
        "flutter",
        "meta",
        "plugin_platform_interface"
      ]
    },
    {
      "name": "win32_registry",
      "version": "1.1.5",
      "dependencies": [
        "ffi",
        "win32"
      ]
    },
    {
      "name": "device_info_plus_platform_interface",
      "version": "7.0.3",
```

```
      "dependencies": [
        "flutter",
        "meta",
        "plugin_platform_interface"
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    },
    {
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      "dependencies": [
        "collection",
        "source_span",
        "string_scanner",
        "typed_data"
      ]
    },
    {
      "name": "sky_engine",
      "version": "0.0.0",
      "dependencies": []
    },
    {
      "name": "material_color_utilities",
      "version": "0.13.0",
      "dependencies": [
        "collection"
      ]
    },
    {
      "name": "characters",
      "version": "1.4.1",
      "dependencies": []
    },
    {
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      "version": "3.1.0",
      "dependencies": [
        "matcher",
        "path",
        "stack_trace",
        "sync_http"
      ]
    },
    {
      "name": "fuchsia_remote_debug_protocol",
      "version": "0.0.0",
      "dependencies": [
        "meta",
        "process",
        "vm_service"
      ]
    },
    {
      "name": "leak_tracker_testing",
      "version": "3.0.2",
      "dependencies": [
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        "matcher",
        "meta"
      ]
    },
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      "version": "11.0.2",
      "dependencies": [
        "clock",
        "collection",
        "meta",
        "path",
        "vm_service"
      ]
    },
    {
      "name": "term_glyph",
```

```
      "version": "1.2.2",
      "dependencies": []
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    {
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      "dependencies": [
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      ]
    },
    {
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      "version": "1.10.1",
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        "path",
        "term_glyph"
      ]
    },
    {
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      "version": "2.1.2",
      "dependencies": [
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        "string_scanner"
      ]
    },
    {
      "name": "plugin_platform_interface",
      "version": "2.1.8",
      "dependencies": [
        "meta"
      ]
    },
    {
      "name": "uuid",
      "version": "4.5.2",
      "dependencies": [
        "crypto",
        "fixnum",
        "meta"
      ]
    },
    {
      "name": "platform",
      "version": "3.1.6",
      "dependencies": []
    },
    {
      "name": "xdg_directories",
      "version": "1.1.0",
      "dependencies": [
        "meta",
        "path"
      ]
    },
    {
      "name": "dbus",
      "version": "0.7.11",
      "dependencies": [
        "args",
        "ffi",
        "meta",
        "xml"
      ]
    },
    {
      "name": "typed_data",
      "version": "1.4.0",
      "dependencies": [
        "collection"
      ]
    },
    {
```

```
      "name": "sync_http",
      "version": "0.3.1",
      "dependencies": []
    },
    {
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      "version": "5.0.5",
      "dependencies": [
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        "path",
        "platform"
      ]
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    {
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      "version": "1.1.1",
      "dependencies": []
    },
    {
      "name": "crypto",
      "version": "3.0.7",
      "dependencies": [
        "typed_data"
      ]
    },
    {
      "name": "xml",
      "version": "6.6.1",
      "dependencies": [
        "collection",
        "meta",
        "petitparser"
      ]
    },
    {
      "name": "args",
      "version": "2.7.0",
      "dependencies": []
    },
    {
      "name": "petitparser",
      "version": "7.0.1",
      "dependencies": [
        "collection",
        "meta"
      ]
    }
  ],
  "configVersion": 1
}
```


File: federated_learning_in_mobile/.dart_tool/flutter_build/dart_plugin_registrant.dart

---

```
//
// Generated file. Do not edit.
// This file is generated from template in file `flutter_tools/lib/src/flutter_plugins.dart`.
//

// @dart = 3.9

import 'dart:io'; // flutter_ignore: dart_io_import.
import 'package:path_provider_android/path_provider_android.dart' as path_provider_android;
import 'package:path_provider_foundation/path_provider_foundation.dart' as path_provider_foundation;
import 'package:battery_plus/battery_plus.dart' as battery_plus;
import 'package:device_info_plus/device_info_plus.dart' as device_info_plus;
import 'package:path_provider_linux/path_provider_linux.dart' as path_provider_linux;
import 'package:share_plus/share_plus.dart' as share_plus;
import 'package:url_launcher_linux/url_launcher_linux.dart' as url_launcher_linux;
import 'package:path_provider_foundation/path_provider_foundation.dart' as path_provider_foundation;
import 'package:device_info_plus/device_info_plus.dart' as device_info_plus;
import 'package:path_provider_windows/path_provider_windows.dart' as path_provider_windows;
import 'package:share_plus/share_plus.dart' as share_plus;
import 'package:url_launcher_windows/url_launcher_windows.dart' as url_launcher_windows;
```

```
@pragma('vm:entry-point')
class _PluginRegistrant {

  @pragma('vm:entry-point')
  static void register() {
    if (Platform.isAndroid) {
      try {
        path_provider_android.PathProviderAndroid.registerWith();
      } catch (err) {
        print(
          '`path_provider_android` threw an error: $err. '
          'The app may not function as expected until you remove this plugin from pubspec.yaml'
        );
      }

    } else if (Platform.isIOS) {
      try {
        path_provider_foundation.PathProviderFoundation.registerWith();
      } catch (err) {
        print(
          '`path_provider_foundation` threw an error: $err. '
          'The app may not function as expected until you remove this plugin from pubspec.yaml'
        );
      }

    } else if (Platform.isLinux) {
      try {
        battery_plus.BatteryPlusLinuxPlugin.registerWith();
      } catch (err) {
        print(
          '`battery_plus` threw an error: $err. '
          'The app may not function as expected until you remove this plugin from pubspec.yaml'
        );
      }

      try {
        device_info_plus.DeviceInfoPlusLinuxPlugin.registerWith();
      } catch (err) {
        print(
          '`device_info_plus` threw an error: $err. '
          'The app may not function as expected until you remove this plugin from pubspec.yaml'
        );
      }

      try {
        path_provider_linux.PathProviderLinux.registerWith();
      } catch (err) {
        print(
          '`path_provider_linux` threw an error: $err. '
          'The app may not function as expected until you remove this plugin from pubspec.yaml'
        );
      }

      try {
        share_plus.SharePlusLinuxPlugin.registerWith();
      } catch (err) {
        print(
          '`share_plus` threw an error: $err. '
          'The app may not function as expected until you remove this plugin from pubspec.yaml'
        );
      }

      try {
        url_launcher_linux.UrlLauncherLinux.registerWith();
      } catch (err) {
        print(
          '`url_launcher_linux` threw an error: $err. '
          'The app may not function as expected until you remove this plugin from pubspec.yaml'
        );
      }

    } else if (Platform.isMacOS) {
      try {
        path_provider_foundation.PathProviderFoundation.registerWith();
```

```
      } catch (err) {
        print(
          '`path_provider_foundation` threw an error: $err. '
          'The app may not function as expected until you remove this plugin from pubspec.yaml'
        );
      }

    } else if (Platform.isWindows) {
      try {
        device_info_plus.DeviceInfoPlusWindowsPlugin.registerWith();
      } catch (err) {
        print(
          '`device_info_plus` threw an error: $err. '
          'The app may not function as expected until you remove this plugin from pubspec.yaml'
        );
      }

      try {
        path_provider_windows.PathProviderWindows.registerWith();
      } catch (err) {
        print(
          '`path_provider_windows` threw an error: $err. '
          'The app may not function as expected until you remove this plugin from pubspec.yaml'
        );
      }

      try {
        share_plus.SharePlusWindowsPlugin.registerWith();
      } catch (err) {
        print(
          '`share_plus` threw an error: $err. '
          'The app may not function as expected until you remove this plugin from pubspec.yaml'
        );
      }

      try {
        url_launcher_windows.UrlLauncherWindows.registerWith();
      } catch (err) {
        print(
          '`url_launcher_windows` threw an error: $err. '
          'The app may not function as expected until you remove this plugin from pubspec.yaml'
        );
      }

    }
  }
}
```

File: federated_learning_in_mobile/.dart_tool/flutter_build/3430039a9863faeb85c025b07c39c70d/dart_build_result.json

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File: federated_learning_in_mobile/.dart_tool/flutter_build/3430039a9863faeb85c025b07c39c70d/native_assets.json

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{"format-version":[1,0,0],"native-assets":{}}

File: federated_learning_in_mobile/.dart_tool/flutter_build/3430039a9863faeb85c025b07c39c70d/outputs.json

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File: federated_learning_in_mobile/.dart_tool/flutter_build/70a32e70eabdcf98e320a667e5114090/dart_build_result.jso

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